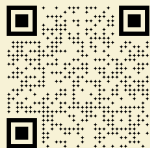


AIEA Lab: eXplainable AI

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Lab: <https://aiea-lab.github.io>

Personal:

<https://biagiomattialarosa.github.io>



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Scenario 1

1 Explainable AI

- You are running late for an important interview
- You have two options:
 1. Follow the standard route and arrive late
 2. Cross a huge, mysterious lawn
- But there is a sign





Scenario 2

1 Explainable AI

- What if the sign is the following?

**Please don't walk on the groundcover.
It's full of snakes.**



Scenario 3

1 Explainable AI

- And now?





ChatGPT - Prompt

1 Explainable AI

- Ask ChatGPT for its opinion about something
 - For example, upload one of your photos and ask how old you were in that photo
 - The question should not concern factual information (i.e., something easily found on Google).
- Done? Was the answer correct?



ChatGPT - Prompt

1 Explainable AI

- My initial prompt: *How old do you think I was in this picture? Give me a 2-3 year range.*



- Input:

Based on your appearance in the photo, I would estimate your age to be between **28**

- Answer: **and 30 years old.**
- Think about our previous scenarios: ChatGPT is writing the “instructions”
- In which scenario are we?



ChatGPT - Explanations

1 Explainable AI

- Now let's ask an explanation to ChatGPT
- Ask for the reasoning behind that specific answer.
- Done? Did the reasoning make sense in relation to the correctness of the answer?
- Think about our previous scenarios: ChatGPT is writing the “instructions”
- In which scenario are we?



ChatGPT - Explanations?

1 Explainable AI

- Think about our previous scenarios: ChatGPT is writing the “instructions”
- In which scenario are we?
 - **We are still in Scenario 1!!**
 - There is no difference (from the AI’s point of view) between your first request and second request
 - The motivation provided is not linked to the actual decision process! It’s just another prediction!



ChatGPT - Explanations?

1 Explainable AI

- New (incognito) session:

We had a previous conversation about this photo where I asked: "how old do you think I was in this picture? Give me a 2-3 years range" and you answered: "Based on your appearance in the photo, I would estimate your age to be between 23 and 25 years old". Do you confirm your guess? Can you provide the motivations that let you arrive to your guess?

Thanks for the follow-up!

Yes, I can confirm the previous age estimate — **23 to 25 years old** — based on the features visible in your photo. Here's the reasoning behind that range:



ChatGPT - Explanations?

1 Explainable AI

- Skin and facial features:
 - **Now:** Your skin appears smooth and relatively free from signs of aging (e.g. wrinkles, sagging, or pronounced under-eye bags), which is more typical in people under 25.
 - **Before:** Your skin appears generally smooth with only minimal signs of aging (like faint lines or texture), which suggests you're past your early 20s but not yet in your mid-30s.
 - **Now:** Your face has a youthful shape — less angular or sunken than what we usually see in people closer to 30.
 - **Before:** The proportions of your face and the definition of your features look like those of an adult who's fully matured but still relatively young.
- Expression and posture:
 - **Before:** While neutral, there's a subtle freshness to the expression and skin tone that's often found in people in their early 20s.
 - **Now:** You project a calm, adult demeanor that's typical of someone with a few years of adult experience, not someone still in their early 20s.



eXplainable AI

1 Explainable AI

- Goal: understanding AI behavior
- Examples:
 - What are the most important factors influencing this prediction?
 - What do I need to change in my input to obtain a different prediction?
 - What did this model learn during the training process?
- **Key Difference:** The explanation **must** be truthful to the model's behavior
 - We usually need a mathematical proof/connection
 - E.g., Altering non-important factors should not change the prediction
 - How strong are these guarantees?
 - Strong for some class of models (e.g., Decision Trees, Logistic Regression)
 - Weak (contextualized) for complex models (e.g., LLM, Deep Neural Networks, etc)



eXplainable AI on Complex AI

1 Explainable AI





eXplainable AI - Why

1 Explainable AI





eXplainable AI - Why

1 Explainable AI

- These scenarios are real, not hypothetical ones! An example?

Bias Behind the Wheel: Fairness Testing of Autonomous Driving Systems

and 3,513 skin tone labels. Our findings reveal significant fairness issues, particularly related to age. The proportion of undetected children is 20.14% higher compared to adults. Furthermore, we explore how various driving scenarios affect the fairness of pedestrian detectors. We find that pedestrian detectors demonstrate significant gender biases during night time, potentially exacerbating the prevalent societal issue of female safety concerns during nighttime out. Moreover, we observe that pedestrian detectors can demonstrate both

- Search for COMPAS's bias and fairness!



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2 AIEA Lab

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AIEA Lab

2 AIEA Lab

- Artificial Intelligence
Explainability Accountability
- Mission: To conduct cutting-edge interpretable and explainable AI research that inspires **curiosity**, while actively explaining **systemic injustices** within and outside of AI and computer science through expanded **empowerment**, alongside **fun** and stimulating **collaborators**.





AIEA Lab - Useful Links

2 AIEA Lab

- Guiding Principles: <https://shorturl.at/X5pKC>
- Website: <https://aiea-lab.github.io/>



AIEA Lab - Research

2 AIEA Lab

- The lab is heterogeneous, covering a lot of topics and domains!
- Commonality: explainable AI
- Some projects:
 - NeuroSymbolic
 - Deep Neural Networks (e.g., LLM) + Symbolic Computation (e.g., Logic)
 - Autograder
 - Create a code autograding framework that can give students personalized feedback
 - XAI for Mental Health



AIEA Lab - Research

2 AIEA Lab

- Robust Autonomous Vehicles
 - make autonomous vehicles safer and more robust by
 - designing interesting driving scenarios
 - implementing AI algorithms that can ensure safety
 - explaining the behaviors in training and post hoc
- Network Pruning
 - How efficient are current models?
 - Can we reduce their carbon footprint?
 - What knowledge do we lose when we compress or prune them?



AIEA Lab - Research

2 AIEA Lab

- Understanding LLM failures
 - When do these models usually fail?
 - Why do they fail?
 - How can we correct them?
- Neuron Alignment and Explanation
 - What do these models learn in terms of knowledge?
 - Do they recognize concepts (e.g., object) as humans do?



AIEA Lab - How To Join

2 AIEA Lab

- Info at: <https://aiea-lab.github.io/join/>
- For Undergrads: Auditor program (almost every quarter)
 - Complete 10 tasks over 10 weeks (the full quarter) to join the lab
 - Every project has its own tasks
 - All the tasks are code based
 - Each project has its own knowledge requirements
- Join the waiting list if you want to know more



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3 Neuron Alignment

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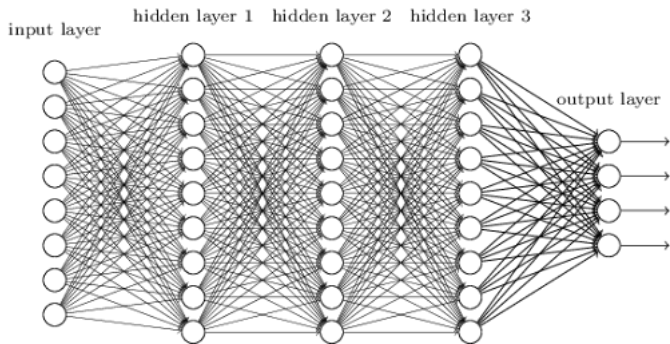
▶ 3 Neuron Alignment



Neuron Alignment

3 Neuron Alignment

- Neurons are the basic functional units of every modern AI
- Each neuron takes some inputs (from previous neurons) and returns an output





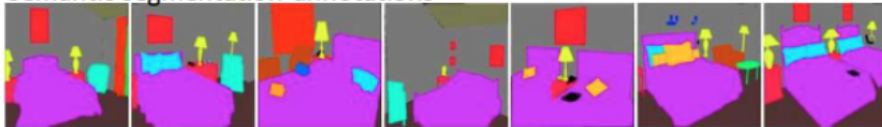
Neuron Alignment

3 Neuron Alignment

- Our goal: understand whether these neurons “parse” inputs similarly to humans.
- In other words: do they see the world as we do?
- But..how do humans identify concepts?



Semantic segmentation annotations



- What about artificial neurons?



Let's do it Together

3 Neuron Alignment

- Tutorial at the following Colab Link: <https://shorturl.at/PkeaN>

The screenshot illustrates the process of changing the Colab runtime configuration. It includes the left-hand menu, the 'Change runtime type' dialog box, and the bottom toolbar. Red boxes and numbers highlight the following steps:

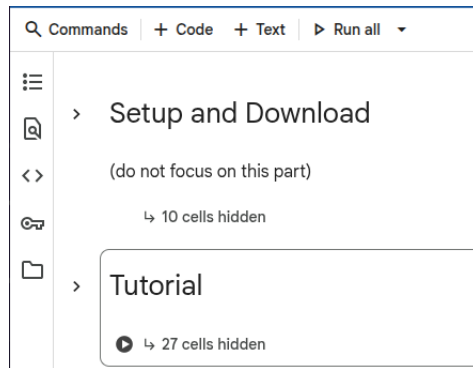
1. Click the 'Reconnect' button in the top left corner of the interface.
2. Click 'Change runtime type' in the left-hand menu.
3. Select 'T4 GPU' under the 'Hardware accelerator' section in the 'Change runtime type' dialog.
4. Click the 'Save' button in the bottom right corner of the 'Change runtime type' dialog.
5. Click the 'Reconnect' button in the bottom right corner of the interface.
6. Click the 'Run all' button in the bottom toolbar.



Tutorial

3 Neuron Alignment

- Two main sections
 - Setup (you can ignore this part)
 - Tutorial





Tutorial

3 Neuron Alignment

```
unit = 10 # ID (number) of the specific neuron we want to analyze  
layer = 'layer4' # Layer where the neuron is placed  
model = 'resnet18' # Model where the layer is placed  
num_clusters = 1 # 1 = only the highest activations  
length_explanation = 1 # only 1 concept per neuron  
device = 'cuda' # Change this to run on CPU to 'cpu'
```

- Parameters
 - unit: Neuron we want to explain



Tutorial

3 Neuron Alignment

- We can skip this part. It creates a structure containing all the configurations needed.

```
# Create Config Object
cfg = settings.Settings(
    subset='ade20k',
    model=model,
    pretrained='places365',
    num_clusters=num_clusters,
    beam_limit=5,
    device=device,
)
```

```
# Variables useful later
mask_shape = cfg.get_mask_shape()
segmentations_directory = cfg.get_segmentation_directory()
```



Tutorial - Loading the dataset

3 Neuron Alignment

- Here we load the images from the datasets and their segmentations

```
# Load the dataset
```

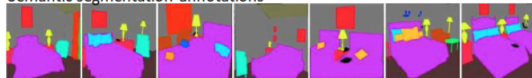
```
image_dataset = segmentations.BrodenDataset(  
    cfg.dir_datasets,  
    subset=cfg.index_subset,  
    resolution=cfg.get_img_size(),  
    broden_version=1,  
    transform_image=torchvision.transforms.Compose([  
        torchvision.transforms.Resize(112)])  
)
```

```
# Load the annotations masks for all the possible concepts
```

```
masks = mask_utils.get_masks(  
    segmentations_directory, None, image_dataset.labels,  
    device  
)
```



Semantic segmentation annotations





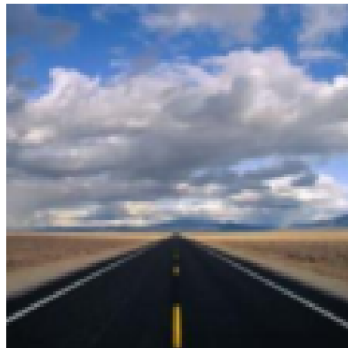
Tutorial - Neurons Output

3 Neuron Alignment

- How does a neuron's output look like?
- Let's consider a random sample from the dataset

```
# ID sample  
sample_id = 2767 # Why 2767? More on this later  
image_sample, _, _ = image_dataset[sample_id]  
vis_image_sample = convert_and_show(image_sample)  
plt.imshow(vis_image_sample)
```

<matplotlib.image.AxesImage at 0x7d8965d8cfb0>





Tutorial - Neurons Output

3 Neuron Alignment

- How does a neuron's output look like?

```
# Load Activations from cache
```

```
layer_dir = f"{cfg.get_activation_directory()}/{layer}"
```

```
unit_activations = torch.from_numpy(np.load(f"{layer_dir}/{uni
```

```
# Print activations on this sample
```

```
print(unit_activations[sample_id])
```

```
[[0.1472, 0.2041, 0.3553, 0.5312, 0.7045, 0.8311, 0.8136],  
 [0.4104, 0.5994, 0.7316, 0.9899, 1.1755, 1.4022, 1.1227],  
 [0.8508, 1.1441, 1.3514, 1.7521, 1.6452, 1.7191, 1.3564],  
 [2.9625, 4.6061, 5.4939, 6.4207, 5.8436, 5.1065, 3.4636],  
 [4.9507, 7.1956, 8.1267, 8.7644, 8.3001, 7.7230, 5.8234],  
 [5.4765, 8.2466, 9.2347, 8.4777, 9.2783, 8.7648, 6.1887],  
 [4.4002, 6.2503, 5.7696, 5.1491, 5.5792, 6.0862, 5.2521]]
```

- It's chaotic, isn't it?



Tutorial - Neurons Output

3 Neuron Alignment

- If we look more carefully, we can notice some spatial patterns

```
[ [0.1472, 0.2041, 0.3553, 0.5312, 0.7045, 0.8311, 0.8136],  
  [0.4104, 0.5994, 0.7316, 0.9899, 1.1755, 1.4022, 1.1227],  
  [0.8508, 1.1441, 1.3514, 1.7521, 1.6452, 1.7191, 1.3564],  
  [2.9625, 4.6061, 5.4939, 6.4207, 5.8436, 5.1065, 3.4636],  
  [4.9507, 7.1956, 8.1267, 8.7644, 8.3001, 7.7230, 5.8234],  
  [5.4765, 8.2466, 9.2347, 8.4777, 9.2783, 8.7648, 6.1887],  
  [4.4002, 6.2503, 5.7696, 5.1491, 5.5792, 6.0862, 5.2521]]
```

```
[ [0.1472, 0.2041, 0.3553, 0.5312, 0.7045, 0.8311, 0.8136],  
  [0.4104, 0.5994, 0.7316, 0.9899, 1.1755, 1.4022, 1.1227],  
  [0.8508, 1.1441, 1.3514, 1.7521, 1.6452, 1.7191, 1.3564],  
  [2.9625, 4.6061, 5.4939, 6.4207, 5.8436, 5.1065, 3.4636],  
  [4.9507, 7.1956, 8.1267, 8.7644, 8.3001, 7.7230, 5.8234],  
  [5.4765, 8.2466, 9.2347, 8.4777, 9.2783, 8.7648, 6.1887],  
  [4.4002, 6.2503, 5.7696, 5.1491, 5.5792, 6.0862, 5.2521]]
```



Tutorial - Our Research

3 Neuron Alignment

- Our goal is to check if these patterns are **aligned** with human-defined concepts (i.e., segmentations)
- The hard part is to find patterns that are **consistent** across multiple samples
- Our research aims to find explanations that tell us whether a neuron's activations are aligned or not and to which concept they are aligned
- Let's do it!



Tutorial - Neurons Activations

3 Neuron Alignment

- Let's focus at first on the highest activations
- We extract a “threshold” that we can use to determine which activations are considered the “highest one”
 - In our case, we select the top 5 percentile
- We use that threshold to binarize the activations (i.e. set all activations below the threshold to 0 and others to 1)

```
# Compute activation range to be kept in the masks
activation_range = activation_utils.compute_activation_range(
    unit_activations, num_clusters) [0]
lower, upper = activation_range
print(activation_range)

(tensor([4.3697], dtype=torch.float64), tensor(inf))
```



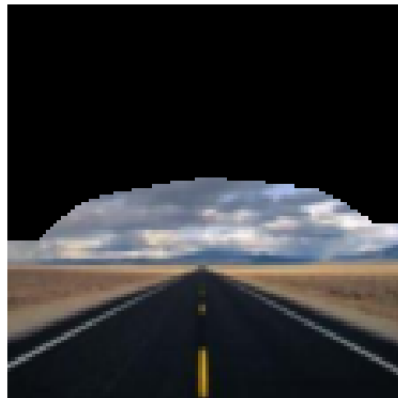
Tutorial - Neurons Activations

3 Neuron Alignment

- Result:

```
binary_activations = torch.where(  
    (unit_activations > lower) & (unit_activations < upper),  
    1, 0)  
print(binary_activations[sample_id])
```

```
tensor([[0, 0, 0, 0, 0, 0, 0],  
        [0, 0, 0, 0, 0, 0, 0],  
        [0, 0, 0, 0, 0, 0, 0],  
        [0, 1, 1, 1, 1, 1, 0],  
        [1, 1, 1, 1, 1, 1, 1],  
        [1, 1, 1, 1, 1, 1, 1],  
        [1, 1, 1, 1, 1, 1, 1]])
```



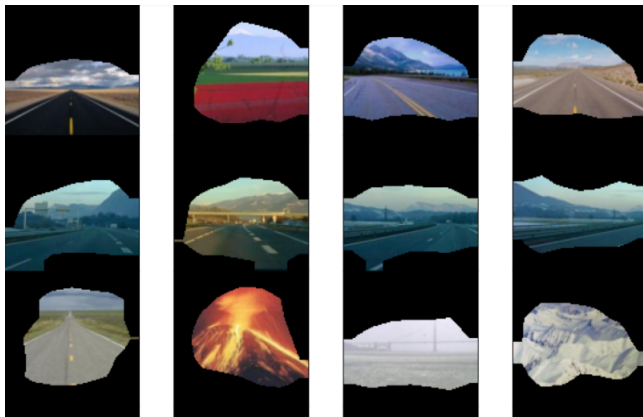
- Is the neuron activation aligned with something?



Tutorial - Neurons Activations

3 Neuron Alignment

- Unfortunately, 1 sample is not enough
- We need to test multiple samples
- From now on, I will just show images, since the code simply loads cached files





Tutorial - Explanations

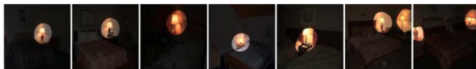
3 Neuron Alignment

- Now it's time to assign labels to these visualized concepts
- How? Network Dissection selects the concept whose human segmentation maps overlap the most with these computed neurons masks

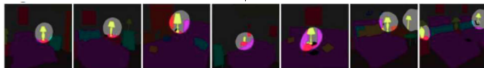
Semantic segmentation annotations



overlapping with



=



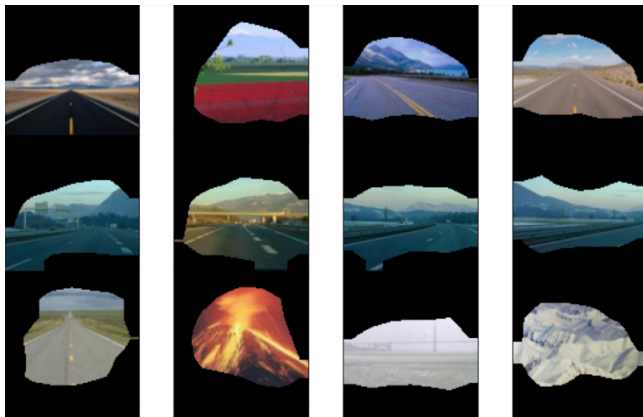
This neuron recognizes the **lamp** concept



Tutorial - Explanations

3 Neuron Alignment

- In this case, Network Dissection returns **Highway** as an explanation
- How good is it?

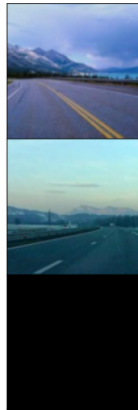




Tutorial - Explanations

3 Neuron Alignment

- In this case, Network Dissection returns **Highway** as an explanation
- How good is it?
- These are the masks annotated by humans as “highway”!
- Do you see any problems?





Tutorial - XAI Scenario

3 Neuron Alignment

- Does this scenario make sense now?

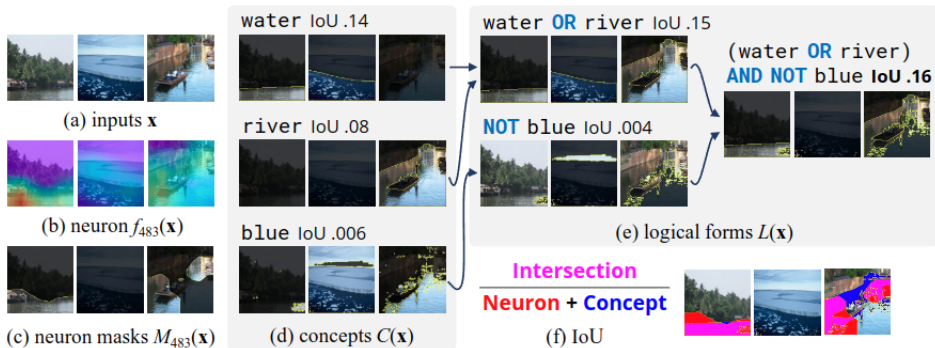




Tutorial - Explanations

3 Neuron Alignment

- Let's try to do better! Compositional Explanations propose to search for the **combination** of concepts up to length n that maximize the overlap
- How? Using search algorithms!





Tutorial - Explanations

3 Neuron Alignment

- Let's set the maximum length to 3
- In this case, Compositional Explanations Algorithm returns **((highway-s OR field-cultivated-s) AND NOT sky)** as an explanation
- How good is it?
- Better than before?





Tutorial - Explanations

3 Neuron Alignment

- ((highway-s OR field-cultivated-s) AND NOT sky)
- How good is it?
- Better than before?
 - It is at the math level (IoU)
 - NetDissect: 0.065
 - Compositional: 0.087





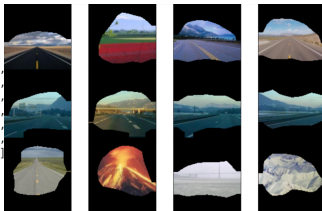
Tutorial - What's Next?

3 Neuron Alignment

- Activation Matrix

```
[[0.1472, 0.2041, 0.3553, 0.5312, 0.7045, 0.8311, 0.8136],  
 [0.4104, 0.5994, 0.7316, 0.9899, 1.1755, 1.4022, 1.1227],  
 [0.8508, 1.1441, 1.3514, 1.7521, 1.6452, 1.7191, 1.3564],  
 [2.9625, 4.6061, 5.4939, 6.4207, 5.8436, 5.1065, 3.4636],  
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 [5.4765, 8.2466, 9.2347, 8.4777, 9.2783, 8.7648, 6.1887],  
 [4.4002, 6.2503, 5.7696, 5.1491, 5.5792, 6.0862, 5.2521]]
```

- Neuron's Mask



- Explanation's Mask





Our Research

3 Neuron Alignment

- Multiple activation ranges
- Replace human annotations with AI
- Dynamic explanation length
- Optimality of explanation
- Application over multiple domains (e.g., LLMs)
- and so on..



Our Research

3 Neuron Alignment



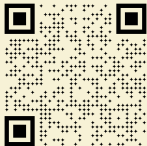
The ultimate goal is to capture as much behavior as possible. (i.e., increase the validity of explanations)



Summary

4 Summary

- Explainable AI
- AIEA Lab
- Neuron Alignment



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